INCH-POUND

A-A-55286
18 SEPTEMBER 1995

COMMERCIAL ITEM DESCRIPTION

CLOTH, POPLIN, COTTON

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-C-507 for all federal agencies.

1. SCOPE. This commercial item description covers the requirements for cotton poplin cloth intended for use in the manufacture of clothing items.

2. SALIENT CHARACTERISTICS

2.1 General description. The cloth shall be a plain weave, woven with cotton which has been carded, combed, drawn, and spun into 2-ply yarns for the warp and single yarns for the filling. The cloth shall be closely singed, preshrunk, desized and mercerized, and shall have a clear, lustrous finish. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket, or package. The width shall be as specified in the contract or purchase order, inclusive of selvages. The use of sulfur dyes and compounds capable of oxidation to sulfuric acid is prohibited. The dyed and finished cloth shall show no more labile sulfur than that shown by the standard sample. When a standard sample is not available the dyed and finished cloth shall show no more than a slight trace of labile sulfur when tested as specified in accordance with Test Method 2020 of FED-STD-191 - Textile Test Methods.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FQSC, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A FSC 8305

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2.2 Physical requirements. The finished cloth shall conform to the physical requirements indicated in Table I.

TABLE I. Physical requirements.

Characteristic	Requirement Min.	Requirement Max.	Test Method
Weight (ozs./sq. yd.)	4.5	5.0	ASTM-D 3776 (Method C)
Yarns per inch			ASTM-D 3775
Warp	106		
Filling	48		
Breaking strength (lbs.)			ASTM-D 5034
Warp	110		
Filling	50		
Discoloration (bleached)	Equal to Standard or Fair		AATCC 16
pH Value	5.5	8.5	2811 1/
Nonfibrous material content		2.0%	ASTM-D 629 (Section 9)
Seam efficiency	85		5110 2/
Shrinkage			AATCC 96
Warp		2.0%	Opt. Vc 3/ 4/ 5/
Filling		2.0%	

1/ Test Methods as specified in FED-STD-191 - Textile Test Methods.
2/ The thread shall conform to type I of A-A-52094. The needle size shall be 0.032 ± 0.001 inch and the top (needle) thread shall be 50/3. The bottom (looper) thread shall be 70/2.
3/ Sample size shall be 22 X 22 inches with 18 inch benchmarks, 6 inches apart.
4/ Use drying procedure (D), or samples may be dried using a thermostatically controlled circulating oven at 221° to 230°F until dried. The sample shall be allowed to cool for 5 minutes, sprinkled with water to permit damp pressing and allowed to stand in this condition for 5 minutes.
5/ The dampened specimens shall be pressed with a flatbed press as described in AATCC 96 Section 5.2 or a hand iron as described in AATCC 96 Section 5.10 at a temperature of 275° to 300°F. Section 7.4 of AATCC 96 shall not apply.

2.3 Color. The color shall be as specified in the contract or purchase order, and shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° K and shall be a good approximation to the standard sample under incandescent lamplight at 2800° K. When white is specified, the cloth shall be chemically bleached and may be supplemented with optical brighteners with fluorescence to the blue region. The degree of whiteness shall match the standard sample.

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2.3.1 Colorfastness. The dyed and finished cloth shall show colorfastness to light, combined laundering and bleaching, perspiration and crocking equal to or better than the standard sample. As a limit of acceptability or when no standard sample is available, the dyed and finished cloth shall show a minimum rating of "4" or (good) colorfastness when tested in accordance with the indicated test methods:

AATCC 16, Option A - Colorfastness to Light with the following exceptions: When a color change equal to Step 4 on the Gray Scale for change in color is evident in either sample being evaluated or the standard sample, exposure shall be continued for an additional number of hours equal to the number of hours at which the color change was apparent. Specimens which do not show a color change until 80 AFU's shall be returned to the machine for a total exposure period not to exceed 140 hours.

AATCC 61, Test 4A - Colorfastness to Combined Laundering and Bleaching with the following exceptions: The soap solution shall contain 0.5 percent soap P-S-1792 - Soap, Laundry, (Neutral and Built), Type 1, classes 1 or 2; and 0.2 percent sodium metasilicate, O-S-604 - Sodium Metasilicate, Technical, Type I. The bleach solution shall have a 1.0 percent of available chlorine. The specimen shall be 4.0 grams.

AATCC 15 - Colorfastness to Perspiration with the following exception: The test for colorfastness to alkaline solution shall also apply.

AATCC 8 - Colorfastness to Crocking with the following exception: The AATCC Chromatic Transference Scale rating shall be used.

2.4 Label/tag. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act. The face side of the cloth shall be identified by stamping the word "Face" at each end of the roll. This label/tag shall be located so that it is completely visible on each roll when it is packaged as specified and so that it causes no damage, and remains on the roll as cloth lengths are distributed.

3. REGULATORY REQUIREMENTS

3.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR) to the maximum extent practicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves

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the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract or purchase order.

4.2 Market acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order. The company offering the cloth put up on rolls, or generically similar cloth as described herein, must have been sold to the commercial market or to the Government for at least 2 years.

4.3 Warranty. The item offered shall include the standard warranty given to the commercial market beginning with the date of delivery of the individual item.

4.4. Visual examination. The cloth shall be examined as indicated in 4.4.1 through 4.4.1.4.

4.4.1 Yard-by-yard examination. Each piece in the sample unit shall be examined on the face side only. When the total yardage in the piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of FED-STD-4 - Glossary of Fabric Imperfections, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on figure 1 of FED-STD-4 shall be scored. No folded linear yard shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following values: 28 points for dyed cloth, and 32 points for white cloth. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following point values: 42 points for dyed cloth, and 48 points for white cloth. If one piece exceeds the point level, a second sample of 20 pieces shall be examined only for individual piece quality examination. The lot shall be unacceptable if one or more pieces in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yds inspected}} = \text{Points per 100 sq.yds.}$$

4.4.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

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Note: The following defects when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth; objectionable odor; width less than specified; poor dye penetration, mottled, streaky, or cloudy; excessive neppiness.

4.4.1.2 Examination for length. During the yard-by-yard examination, each piece shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.4.1.3 Examination for shade. During the yard-by-yard examination, each piece shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.1.4 Examination for identification of preshrinkage process, compliance with Textile Fiber Products Identification Act and face marking. During the yard-by-yard examination, each piece in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample: Preshrinkage process identified by name or trademark on cloth or ticket; not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act; face stamping missing from either or both ends; face stamping on wrong side.

5. PACKAGING

5.1 Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

6.1 Sources of Government documents. Copies of military and Federal documents are available from:

Department of the Navy
Defense Printing Service
Detachment Office
700 Robbins Avenue, BLDG 4D,
Philadelphia, PA 19111-5093

Rules And Regulations Under The Textile Fiber Products Identification Act - Application for copies may be addressed to:

Federal Trade Commission
Washington, DC 20580

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6.1.1 Sources of non-Government documents

AATCC Test Methods - Applications for copies should be addressed to:

American Association of Textile Chemists and Colorists
P.O. Box 12215
Triangle Park, NC 27709-2215

ASTM Test Methods - Applications for copies should be addressed to:

American Society For Testing and Materials
1916 Race Street
Philadelphia, PA 19103-1187

MILITARY INTERESTS:

Custodians
Army - GL

Review Activity:
Army - MD

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

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